

Mohit Kumar Dey

Software Engineer | AI, Automation & Full-Stack Development

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SUMMARY

Software Engineer with foundational experience in AI & Full-Stack Development building at the intersection of large language models and production software. Experienced designing RAG pipelines, LLM-integrated workflows, and agentic automation using LangChain, LangGraph, and modern model APIs, alongside full-stack development in React, Next.js, and Node.js. Focused on turning fast-moving AI capabilities from retrieval-augmented generation to autonomous agents into scalable, reliable, user-facing products.

EDUCATION

Chandigarh University

Bachelor of Engineering in Computer Science Engineering

2021 – 2025

CGPA: 8.08

EXPERIENCE

Freelance AI & Full Stack Developer

Nov 2025 – Present

Remote

- Developed AI-powered web applications and automation solutions for clients using Python, React, Next.js, Node.js, and LLM APIs, reducing manual client effort by up to ~85%.
- Delivered responsive business websites and integrated AI-assisted workflows to improve productivity through streamlined operations.
- Designed and implemented scalable REST APIs, backend services, and PostgreSQL databases, maintaining ~99% uptime for critical backend services.
- Collaborated directly with clients to gather requirements, deliver features, and provide ongoing technical support with ~90% positive feedback

Matrice AI - AI Engineer

Aug 2025 – Nov 2025

Remote

- Faced with 1,000+ daily operational events requiring manual handling, constructed automated integrations across messaging, mapping, and task-management APIs - **cutting manual effort by ~90%**.
- To raise system throughput, designed Python/LangChain automation workflows and configured workflow orchestration tooling **increasing processing throughput by 40%**.
- When incident response was slow, executed an event-driven microservice routing with rule-based escalation logic - **cutting resolution time 45% while sustaining 99.9% uptime**.
- To catch failures before they reached production, established monitoring and observability pipelines **keeping 95th-percentile response times under 2 minutes**.

Vensure Employer Solutions - Data Engineer

Sep 2024 – Jun 2025

Noida

- Tasked with processing 50k+ daily EDI records (834, CSV, Fixed Width), engineered automated pipelines in C#, Python, and SQL - **achieving 100% data accuracy and 95% test coverage**.
- To reduce file-transfer downtime, configured and orchestrated secure SFTP workflows using Python, WinSCP, and Jenkins, **improving reliability 35% and cutting downtime 28%**.
- When defect rates threatened SLA targets, deployed regex-based mapping logic and automated QA validations **reducing defects 22% and improving SLA delivery 30%**.
- Collaborated with cross-functional teams to align pipelines with enterprise architecture **sustaining 99% data delivery compliance**.

PERSONAL PROJECTS

AI-Powered Event Organizer | [LiveZ](#)

Jul 2025 – Jan 2026

- Developed a full-stack AI-enabled event management platform using Next.js, Tailwind CSS, and shadcn/ui, delivering responsive, production-ready user experiences.
- Architected secure backend workflows with Convex and Clerk authentication, enabling real-time data handling and protected routes.
- Integrated LLM-based features using the Gemini API to automate content generation and enrich event data.

AI-Enhanced Fraud Detection Pipeline | [LiveZ](#)

Feb 2025 – Jun 2025

- Engineered a scalable fraud-detection pipeline on transaction data using Python and Spark, with PostgreSQL storage.
- Trained ML/DL models (TensorFlow, PyTorch) and served them via Flask APIs in Docker containers on AWS.
- Implemented a RAG pipeline with LangChain and the Gemini API for similarity-search-based anomaly detection, orchestrating async scoring with RabbitMQ, **reaching 95% detection accuracy with sub-second response times.**

Real-Time Cryptocurrency Tracker | [LiveZ](#)

Jan 2024 – May 2024

- Built a real-time full-stack crypto tracking platform using React 18, TypeScript, Express.js, PostgreSQL, and WebSockets for live price updates, watchlists, and alerts.
- Architected scalable REST APIs and real-time data pipelines integrating third-party crypto APIs with secure authentication.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C++

AI & ML: LangChain, LangGraph, LangSmith, LLM APIs, RAG Pipelines, Vector Databases, Prompt Engineering, Agentic AI, Embeddings, TensorFlow, PyTorch

Backend & APIs: Node.js, Nest.js, Express, FastAPI, Flask, RESTful APIs, Microservices Architecture, WebSockets

Databases & Data: PostgreSQL, MongoDB, Redis, RabbitMQ, ETL, Data Pipelines, Apache Spark

Frontend & UI: React, Next.js, HTML5, Tailwind CSS, Framer Motion, Three.js, Figma, GSAP

Cloud & DevOps: AWS (EC2, Lambda, S3), Docker, CI/CD (Jenkins), Postman, Git

Automation & Testing: n8n, Make, Selenium, BeautifulSoup, Playwright, Clay, Apollo

Methodology: Agile, Client-Facing Solutions Delivery

ACHIEVEMENTS

- Achieved Gold Medal (Top 5%) in NPTEL's all-India Internet of Things examination, organized by IIT Kharagpur.
- Awarded Intern of the Month (February 2025) for successfully automating 3 internal workflows.

CERTIFICATIONS

- AWS Certified Cloud Practitioner - AWS
- Introduction to Artificial Intelligence (AI) - IBM, Coursera
- Data Structures and Algorithms - University of California San Diego, Coursera
- Networking Basics - Cisco Networking Academy
- Network Support and Security - Cisco Networking Academy